
Click-Through Rate Prediction on E-Commerce Advertising Data: An Empirical Study of Classical and Deep Learning Approaches

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Abstract

Click-through rate (CTR) prediction is an important task in online advertising, where even small improvements in accuracy can lead to substantial revenue gains. Most existing comparative studies rely on aggregate measures such as AUC and LogLoss, leaving the questions of where and why models differ in practice unresolved. We bridge this gap with a multi-dimensional comparison of five CTR models—LR, LightGBM, Wide & Deep, DeepFM, and DIN—on the Alibaba Taobao Display Advertising Dataset. Beyond overall performance, we analyze cross-model agreement, failure modes, attention behavior, and sample complexity. The AUC progression from LR (0.5779) to DIN (0.6235) reflects four sources of representational gain with diminishing returns at the top. All models miss around 60% of clicked impressions, which concentrate on cold-start users. DIN’s attention mechanism favors semantic relevance over recency, and classical and deep models capture different subsets of clickers, suggesting that ensembling has untapped potential. Together, these results suggest that the remaining performance headroom is representational rather than data-driven: future gains will likely come from richer signals for cold-start users rather than from deeper architectures alone.

1. Introduction

In industrial settings, advertising is the primary revenue source for internet companies. The effectiveness of ad monetization depends critically on click-through rate (CTR) prediction, which directly drives ad ranking and delivery. Even a 0.1% improvement in prediction accuracy can translate into millions of dollars in additional revenue (Cheng et al., 2016).

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Despite this importance, existing comparative studies either evaluate models on datasets lacking behavioral sequence data, or restrict their analysis to aggregate metrics such as AUC and LogLoss, leaving open the question of *where* and *why* models differ in practice.

In this work, we address this gap by conducting a multi-dimensional comparative analysis of five CTR prediction models on the Alibaba/Taobao Display Advertising dataset, which natively supports user behavior sequence modeling.

Contributions.

- We benchmark five CTR models — Logistic Regression, LightGBM, Wide & Deep, DeepFM, and DIN — on the public Alibaba/Taobao Display Advertising dataset.
- We conduct a stratified analysis separating cold-start users (<10 impressions) from active users (≥10 impressions) to quantify the impact of data sparsity.
- We present sample complexity curves showing model performance as a function of training set size, and analyze the failure modes of the best-performing model.

2. Related Work

2.1. CTR Prediction: A Brief Survey

CTR prediction has evolved through a series of architectural advances, each addressing a distinct limitation of its predecessor (Cheng et al., 2016; Guo et al., 2017; Zhou et al., 2018). Early systems relied on logistic regression with manually engineered feature crosses (McMahan et al., 2013), which proved computationally efficient but limited in expressive power. Tree-based methods such as LightGBM (Ke et al., 2017) introduced nonlinear feature interaction without requiring neural architectures. The shift to deep learning brought hybrid designs such as Wide & Deep (Cheng et al., 2016) and end-to-end feature interaction models such as DeepFM (Guo et al., 2017). More recently, attention-based architectures such as DIN (Zhou et al., 2018) moved beyond feature interaction to explicitly model the diversity and locality of user interests from behavioral history. We discuss our rationale for selecting these five models as evaluation targets in Section 4.

2.2. Existing Work on the Taobao Dataset

The Alibaba Display Advertising Dataset (Alibaba Group, 2018) has been used in a small number of comparative studies. Yang et al. (2025) evaluate multiple CTR models on this dataset—including LR, LightGBM, and several deep architectures—and report aggregate metrics such as AUC and LogLoss. However, their analysis stops at model-level performance comparison and does not examine *where* or *why* models differ: which users or impressions drive the gap, what errors persist in the best model, or how models behave under data sparsity. More broadly, standard benchmarking efforts such as Zhu et al. (2021) conduct rigorous multi-model comparisons but rely on anonymized datasets (Criteo, Avazu) that lack explicit user behavior sequences, making them unsuitable for evaluating attention-based models such as DIN under realistic e-commerce conditions.

2.3. Positioning of This Work

This work addresses that gap. We focus on the e-commerce advertising scenario and evaluate models spanning the full spectrum from classical linear approaches (LR) to attention-based deep learning architectures (Wide & Deep, DeepFM, DIN) on the Alibaba Display Advertising Dataset (Alibaba Group, 2018), which natively supports user behavior sequence modeling. Beyond standard metrics, our analysis includes datapoint-level prediction agreement across models, failure mode characterization, and fine-grained behavioral breakdowns — providing a richer understanding of where and why models differ in the e-commerce CTR prediction setting.

3. Data

3.1. Source and Schema

The dataset we use is the public benchmark for click-through rate (CTR) prediction given by Alibaba, the Alibaba Taobao Display Advertising dataset (Alibaba Group, 2018). This dataset contains impression records of eight days from the display advertising system of Taobao for the period of May 5th to May 13th, 2017. Each record represents an ad impression with the user’s click feedback. The aim is to forecast whether a person will click on an ad, given user, ad and contextual variables.

The data collection contains four relational tables. The `raw_sample` table provides impression-level click labels that are utilized for supervised training. `ad_feature` has ad-side metadata, such as category, campaign and pricing. `user_profile` has demographic features about the user. `behavior_log` contains the history actions of the user such as page views, favorites, add-to-cart and buy.

We left join `ad_feature` and `user_profile` to

`raw_sample` as static features. `behavior_log` is used for building the user behavior sequences for sequence-aware models.

3.2. Dataset Statistics

The dataset consists of 26.5M labeled impressions gathered over 8 days covering 1.14M users, 846K advertisements, and 6,769 product categories. The test-set click-through rate is 5.04%, which is representative of the strong class imbalance frequent in display advertising tasks.

There is a distinct long-tail distribution of user activity. The median user only sees 7 impressions, the mean is 23.3 and the 99th percentile reaches 286. Most impressions come from the top 10% most active users, leaving a mix of highly active and cold-start users in the same dataset.

Traffic volumes also vary widely across the day. Impression volume peaks around 22:00 with over 1.9M impressions per hour and drops to around 134K impressions per hour around 04:00, a difference of nearly 14×. Despite this variance, the CTR remains very consistent throughout the day at about 5.0–5.2%.

The behavior history table is huge with 723M interaction events from 1.13M users with non-empty logs. The dataset also has several patterns of missing values: 5.8% of impressions belong to users missing from `user_profile`; 29.1% of ads have missing `brand`; and 54% of users have missing `pvalue_level`.

3.3. Cleaning and Preprocessing

Before training we apply several preprocessing steps. First, we remove price values greater than 500,000 RMB to drop obvious placeholder values such as 99,999,999 (less than 0.01% of ads). We also use a $\log(\text{price} + 1)$ transformation to reduce skewness.

Second, corrupted timestamps beyond the normal range of the dataset are dropped from `behavior_log`. We also drop the `brand` feature since it is extremely sparse, with most brands appearing only a few times.

The missing values in column `pvalue_level` and column `new_user_class_level` are replaced by a reserved value of 0. For impressions whose users are missing from `user_profile`, the missing profile fields are filled with 0 and a binary indicator `has_profile` is added. We do not discard these impressions as the empirical CTR (0.0518) is practically the same as the global CTR (0.0519), meaning that they do not differ in behavior from the whole population.

3.4. Feature Engineering

Raw Unix timestamps are converted to Beijing time (UTC+8), from which we extract hour-of-day and day-of-week features. High-cardinality categorical IDs are mapped to contiguous integer indices for embedding-based models.

In addition to raw features, we compute several user- and ad-level aggregate statistics from the training split only, including per-user, per-ad, and per-(user, category) empirical click rates (`user_ctr`, `ad_ctr`, `user_cate_ctr`), impression counts (`user_imp_count`, `ad_imp_count`), and per-user behavior-log aggregates such as `user_total_pv`, `user_total_buy`, `user_nunique_cates`, and `user_buy_rate`. Low-frequency rates are smoothed toward the global rate using a Bayesian prior:

$$\hat{r} = \frac{n_+ + \alpha r_{\text{global}}}{n + \alpha}, \quad (1)$$

with $\alpha = 10$, which pulls high-variance estimates (e.g., one impression with one click) toward the population mean while preserving accurate estimates for high-volume groups.

3.5. Train/Validation/Test Split

To better emulate real-world deployment, we adopt a strict time-based split without shuffling. The training data spans 2017-05-05 to 2017-05-11, validation is 2017-05-12, and the test data is the first half of 2017-05-13. We do not randomly split to avoid leakage from future user behavior.

All aggregate statistics are calculated only on the training split and joined to validation and test as features.

3.6. Behavior Sequence Construction

For DIN, we construct per-impression user behavior sequences from `behavior_log`. For each impression, we retrieve the user’s 50 most recent interacted categories that are strictly before the impression timestamp. Shorter sequences are padded with a reserved zero index.

The strict “before” cutoff guarantees that no future behavior leaks into the model input, which is consistent with the temporal split used during training and evaluation. Approximately 2.1% of impressions are linked to users with no previous history, resulting in an all-zero sequence.

4. Methods

We compare five models that span four sources of representational capacity: nonlinearity, learned embeddings, automatic feature interaction, and user-interest modeling via attention. Each model is introduced below alongside the rationale for its inclusion in the comparison.

4.1. Logistic Regression (LR)

serves as our interpretable lower-bound baseline. Given a feature vector $\mathbf{x} \in \mathbb{R}^d$, it predicts $\hat{y} = \sigma(\mathbf{w}^\top \mathbf{x} + b)$, where σ is the sigmoid function.

4.2. LightGBM

([Ke et al., 2017](#)) is a gradient boosting framework that builds an ensemble of decision trees sequentially, where each tree corrects the residual errors of the previous ones. Unlike standard GBDT implementations that grow trees level-wise, LightGBM adopts a leaf-wise growth strategy, always splitting the leaf with the highest loss reduction. Two key innovations underpin its efficiency on large-scale sparse data: Gradient-based One-Side Sampling (GOSS), which down-samples low-gradient instances while retaining the most informative ones; and Exclusive Feature Bundling (EFB), which reduces effective feature dimensionality by bundling mutually exclusive sparse features. We include LightGBM as our nonlinear non-neural baseline: the gap between it and LR isolates the contribution of nonlinear feature interaction, before any embedding-based machinery is introduced.

4.3. Wide & Deep

Wide & Deep ([Cheng et al., 2016](#)) jointly trains a *wide* linear model and a *deep* feed-forward network. The wide part memorizes specific feature co-occurrences through cross-product transformations; the deep part learns dense embeddings of sparse categorical features that generalize across the input space. The final prediction is:

$$P(Y = 1 \mid \mathbf{x}) = \sigma\left(\mathbf{w}_{\text{wide}}^\top [\mathbf{x}, \phi(\mathbf{x})] + \mathbf{w}_{\text{deep}}^\top \mathbf{a}^{(l_f)} + b\right), \quad (2)$$

where $\phi(\mathbf{x})$ are cross-product feature transformations and $\mathbf{a}^{(l_f)}$ is the final hidden activation of the deep component. We include Wide & Deep as the first neural model in our lineup: the gap between it and LightGBM reflects what learned ID embeddings add over tree-based nonlinearity alone — something tree models cannot naturally produce.

4.4. DeepFM

DeepFM ([Guo et al., 2017](#)) employs the same two-tower structure as Wide & Deep ([Cheng et al., 2016](#)) but substitutes the manually built wide component with a Factorization Machine (FM) ([Rendle, 2010](#)). Instead of hand-crafted cross features, the FM layer automatically learns pairwise feature interactions from the shared embeddings.

The FM component computes a linear term plus pairwise

interactions of feature embeddings:

$$y_{\text{FM}} = w_0 + \sum_{i=1}^m w_i x_i + \sum_{i=1}^m \sum_{j=i+1}^m \langle \mathbf{e}_i, \mathbf{e}_j \rangle x_i x_j, \quad (3)$$

where $\mathbf{e}_i \in \mathbb{R}^k$ is the embedding vector of feature field i , and the interaction term between two fields is expressed as the inner product of their embeddings. The final prediction sums y_{FM} and a deep neural network output y_{DNN} operating on the same embeddings, before passing through a sigmoid.

In practice, this means that DeepFM simultaneously learns low-order and high-order feature interactions within a single architecture. One reason for the inclusion of DeepFM in our study is to test whether automatically learned interactions can be competitive with manually crafted feature crosses in Wide & Deep, or whether domain knowledge in feature engineering still provides a considerable advantage.

4.5. Deep Interest Network (DIN)

Wide & Deep and DeepFM both employ a fixed feature vector to represent each user. That is, the same user representation is used regardless of which candidate ad is being assessed. This assumption is convenient but ignores the fact that only a subset of a user’s history may be relevant to a specific ad.

DIN (Zhou et al., 2018) relaxes this assumption by using a local activation unit to re-weight the historical behaviors dynamically based on the candidate ad. For each impression, DIN constructs a user-interest representation by pooling the user’s prior behavior embeddings with attention weights conditioned on the candidate ad embedding:

$$\mathbf{v}_u(a) = \sum_{t=1}^T \alpha_t(\mathbf{e}_t, \mathbf{e}_a) \mathbf{e}_t, \quad (4)$$

where $\mathbf{e}_a$ is the candidate-ad embedding, $\{\mathbf{e}_t\}_{t=1}^T$ are the user’s behavior-sequence embeddings, and α_t is an attention score produced by a small feed-forward network that takes the behavior embedding, the candidate-ad embedding, their elementwise product, and their difference as inputs. In our setup, the behavior sequence consists of the user’s 50 most recent interacted category IDs. Unlike the standard attention mechanism, DIN does not apply softmax normalization, so the attention weights are not constrained to sum to one.

The resulting user-interest vector is concatenated with the user profile and context features and then fed into a multi-layer perceptron. Following the original DIN paper (Zhou et al., 2018), we adopt Dice activations and mini-batch aware L_2 regularization.

Among our models, only DIN directly consumes the per-impression behavior sequence. Comparing DIN against

DeepFM therefore helps isolate the impact of attention-based interest modeling beyond strong static-feature baselines.

5. Evaluation Framework / Experimental Setup

We evaluate all models on three metrics:

- **AUC** (Area Under the ROC Curve): measures the model’s ability to rank clicked ads above non-clicked ones, regardless of calibration. Range: $[0, 1]$; higher is better, with 0.5 corresponding to random ranking.
- **LogLoss** (Binary Cross-Entropy): penalizes poorly calibrated probability estimates. Range: $[0, \infty)$; lower is better, with 0 indicating perfect predictions.
- **GAUC** (Group AUC): AUC computed per user and averaged weighted by impression count, better reflecting production ranking quality. Range: $[0, 1]$; higher is better, with the same interpretation as AUC but controlling for between-user difficulty variation.

Accuracy is excluded because the dataset is highly imbalanced (most impressions are non-clicks), making it uninformative.

5.1. Training Configuration

Neural models are implemented with DeepCTR-Torch (Shen, 2017) and trained on Apple MPS (Metal Performance Shaders) with CPU fallback for unsupported ops.

Logistic Regression. LR is trained via `SGDClassifier` from scikit-learn with logistic loss and L2 regularization ($\alpha = 10^{-6}$, 3 passes over the training data). Low- and medium-cardinality categorical features are one-hot encoded (yielding $\sim 9\text{K}$ dimensions); high-cardinality IDs (user, adgroup_id, campaign_id, customer) are represented via pre-computed CTR aggregates to avoid an intractably sparse feature matrix. Continuous features are standardized with zero mean and unit variance.

LightGBM. LightGBM is trained with `num_leaves=255`, `learning_rate=0.05`, `feature_fraction=0.8`, and `bagging_fraction=0.8` (frequency 5). Low- and medium-cardinality features are passed as native LightGBM categoricals; high-cardinality IDs are passed as raw integers alongside CTR aggregates. Early stopping is applied with a patience of 50 rounds on validation AUC, over a maximum of 2000 boosting rounds.

Wide & Deep, DeepFM, and DIN. The three neural models share the same core configuration: embedding dimension 16 for all sparse features, MLP hidden units (256, 128), dropout 0.5, Adam optimizer ($\text{lr} = 10^{-3}$), batch size 4096,

and up to 10 epochs with early stopping (patience 2) on validation AUC. The best checkpoint is reloaded for test evaluation.

A key design decision concerns feature asymmetry across models. CTR aggregate features (`user_ctr`, `ad_ctr`, `cate_ctr`, `user_cate_ctr`) are included for LR and LightGBM but excluded for the three neural models. When high-cardinality ID embeddings and per-entity CTR statistics derived from the same training labels are used together, the model can effectively memorize per-entity click rates, causing severe within-train label leakage (train AUC > 0.9, val AUC < 0.6). Tree-based and linear models do not suffer from this issue because they do not learn continuous ID representations jointly with label-derived statistics.

DIN additionally takes a user behavior sequence as input: the last $K = 50$ interacted `cate_ids`, zero-padded, with sequence length passed explicitly. The behavior sequence embedding shares weights with the candidate `cate_id` embedding, consistent with the original DIN formulation (Zhou et al., 2018).

6. Results

6.1. Main Results

Table 1 reports test-set AUC, LogLoss, and GAUC for all five models. Our five-model spectrum spans four sources of representational capacity, and the AUC ordering tracks that progression exactly: DIN at the top (0.6235), followed by DeepFM (0.6211), Wide & Deep (0.6123), LightGBM (0.5973), and LR (0.5779). The 0.046 AUC gap between the strongest and weakest models is substantial for CTR prediction, where 0.001 AUC routinely translates into measurable production lift (Cheng et al., 2016; Zhou et al., 2018).

The progression from LR to LightGBM to Wide & Deep to DeepFM to DIN isolates four sources of gain: nonlinearity (LR $\rightarrow$ LightGBM, +0.019), learned embeddings (LightGBM $\rightarrow$ Wide & Deep, +0.015), automatic feature interaction (Wide & Deep $\rightarrow$ DeepFM, +0.009), and attention-based user interest modeling (DeepFM $\rightarrow$ DIN, +0.002). The diminishing returns suggest that the easy gains come from nonlinearity and embeddings, while sequence-aware modeling contributes a smaller but non-trivial increment.

GAUC values, which measure per-user ranking quality, are uniformly lower than dataset-level AUC but preserve the same model ordering. This consistency indicates that our model ranking is robust to the choice of aggregation scheme.

Last but not least, LogLoss tells a more nuanced story. LightGBM achieves the second-best LogLoss (0.1985), beating both Wide & Deep (0.2067) and DeepFM (0.2014) despite trailing them on AUC. This split between ranking quality (AUC) and probabilistic calibration (LogLoss) antici-

Table 1. Test-set performance of all five models. AUC and GAUC measure ranking quality; LogLoss penalizes miscalibrated probabilities. Best result per metric in bold.

Model	AUC	LogLoss	GAUC
LR	0.5779	0.2552	0.5382
LightGBM	0.5973	0.1985	0.5439
Wide & Deep	0.6123	0.2067	0.5451
DeepFM	0.6211	0.2014	0.5504
DIN	0.6235	0.1957	0.5525

pates the calibration analysis in Section 7.2: LightGBM produces conservative, well-calibrated predictions concentrated near the global mean, while embedding-based models push scores into a broader range that improves discrimination but slightly hurts calibration.

6.2. Cold-Start vs. Active Users

To examine whether the model ranking holds across user populations of different signal density, we stratify all five models’ test AUC by user activity (number of historical impressions) and visualize the segment $\times$ model matrix as a heatmap (Figures 1 and 13).

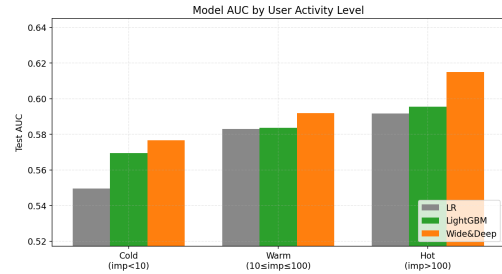


Figure 1. Test-set AUC of all five models stratified by user-activity bucket.

Three patterns emerge. **First**, the relative ranking $LR < LightGBM < Wide \& Deep < DeepFM < DIN$ persists across all activity segments—no architecture dominates only a niche subpopulation, and the spectrum of representational capacity translates into real performance gains for every user type. **Second**, the gap between deep and classical models widens on high-activity users: DIN’s advantage over LightGBM is largest among users with the longest histories, where richer per-user signal allows learned embeddings to specialize, and shrinks substantially on cold-start users where all models converge toward the global CTR. **Third**, every model underperforms sharply on the lowest-activity segment, with AUCs dropping well below their dataset-wide values. This bucket alone contributes a disproportionate share of the residual error characterized in Section 7.2, identifying user-side cold start as the most actionable target for future improvement.

6.3. Sample Complexity

To check whether the gap between classical baselines and deep models can be closed simply by training on more data, we retrain LR and LightGBM on six fractions of the training set and plot their test AUC against training-set size (Figure 2).

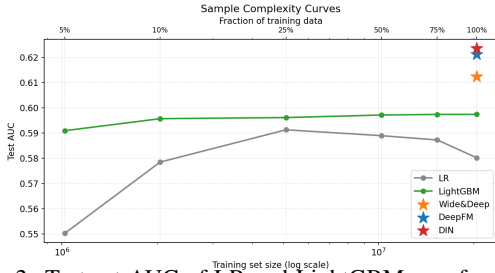


Figure 2. Test-set AUC of LR and LightGBM as a function of training fraction. Both classical baselines saturate well below deep network at full data, indicating that the remaining headroom is representational rather than data-driven.

Within these two models, LightGBM saturates around 25–50% of the training data, gaining only around 0.001 AUC gain from doubling its training size beyond that point. LR’s curve is non-monotonic: AUC peaks at 25% (0.5912) and degrades to 0.5802 at 100%, indicating that its limited representational capacity cannot leverage additional data and begins to overfit narrow feature combinations.

Both classical baselines saturate well below the deep models’ performance at 100% data. Due to the computational cost of retraining neural models across multiple data fractions, this analysis is restricted to LR and LightGBM; the deep models’ full-data performance is shown as reference datapoints for comparison. The remaining headroom therefore does not come from training-set scale but from *representational capacity*: as shown in Figure 2, neither LR nor LightGBM approaches the performance of deep models even at full data, suggesting that the bottleneck is architectural rather than data-driven.

7. Analysis

7.1. Datapoint-Level Agreement Analysis

We first binarize each model’s predictions by applying the top-10% threshold, which corresponds to a positive rate roughly $2\times$ the global CTR. We then count, for each test impression, how many models classify it as a click.

Where do the models agree? Among the 143,527 actual clicks in the test set, only 1.8% are correctly identified by all five models (*easy* clicks), while 59.9% are missed by every model (*hard* clicks). Precision rises monotonically with agreement level, from 4.1% for zero-agreement impressions to 14.7% for full-agreement ones — roughly three times the global CTR. This indicates that cross-model agreement is it-

self a powerful signal of predictive confidence (see Figure 8 in the Appendix for the full recall and precision breakdown by agreement level).

Per-sample difficulty. We additionally quantify per-sample difficulty as the standard deviation of predicted probabilities across the five models (Figure 7). The variance of predictions on clicked impressions is much higher than the variance on non-clicked ones, which implies that positive samples tend to sit near the cross-model disagreement boundary, where additional modeling capacity or richer features could plausibly tip the prediction. The non-click distribution is sharply concentrated at low variance, while the clicked distribution has a long right tail. In practice, a substantial share of true clicks remain genuinely contested even across diverse model architectures. Interestingly, ensemble AUC is lowest not at the highest difficulty scores but in the mid-range ($\sigma \approx 0.02\text{--}0.05$), suggesting that moderately uncertain samples are collectively harder for the ensemble than maximally uncertain ones.

Easy vs. hard clicks. We compare “easy” clicks, which are caught by all five models, with “hard” clicks, which are missed by every model (Figure 9). Easy clicks are typically generated by users with high historical CTR clicking on ads with relatively strong click performance. Hard clicks, in contrast, come from low-activity users clicking on low-CTR ads, where neither user-side nor ad-side history provides a strong enough signal for any model to fire confidently. Since hard clicks make up roughly 60% of all positive labels, improving performance on this slice likely requires signals beyond historical CTR averages, such as content embeddings or broader behavioral context.

Pairwise agreement structure. The pairwise recall matrix (Figure 6) reveals two distinct clusters of models. Classical baselines (LR and LightGBM) form one cluster with substantial mutual agreement, while the deep models (DeepFM and DIN) form a second cluster with relatively high mutual agreement of their own. Agreement between the two clusters is markedly lower, indicating that classical and deep models capture different subsets of click behavior — a structural divergence that persists regardless of overall performance differences between the two families.

7.2. Failure Mode Analysis

Calibration. The reliability diagrams (Figure 10) show that all models are reasonably well calibrated at low predicted probabilities, but underpredict positives at higher predicted probabilities. LightGBM produces the most conservative predictions, with probabilities concentrated near the global CTR—this conservative tendency is probably what drives its competitive LogLoss but middling AUC (Section 6.1). LR is the most overconfident on a small number of high-score samples. DeepFM and DIN produce the broadest dynamic

range while remaining well-calibrated up to a predicted probability of around 0.15.

AUC by feature decile. We also examine DIN by evaluating test AUC across deciles of numerous important variables (Figure 11). The best performance is in the top decile of `user_ctr` where DIN produces an AUC of 0.64, well above the overall test AUC of 0.62. By comparison, AUC decreases to 0.56 in the bottom decile, meaning that the model does substantially worse in cold-start conditions when the users have very limited click history. Also, `user_imp_count` seems to show a similar pattern. The most active users get an AUC of 0.63 with DIN but it drops to 0.56 for the least active decile.

False-negative rate by user activity. DIN’s false-negative rate rises sharply for users with fewer than about 10 historical impressions (Figure 12). These low-activity users overlap heavily with the “hard click” subset in Section 7.1 and account for the universal AUC drop across all five models in Section 6.2. User-side cold start is the dominant remaining source of error.

7.3. Attention Analysis (DIN)

To better understand DIN’s attention mechanism, we extract attention weights α_t from Equation (4) for 5,000 test impressions. Averaged across all impressions, the attention distribution shows no systematic recency bias: the oldest 25 positions receive attention comparable to the newest 25 (a new-to-old ratio of 1.00 $\times$), with observed variation attributable to noise rather than a directional trend (Figure 3a).

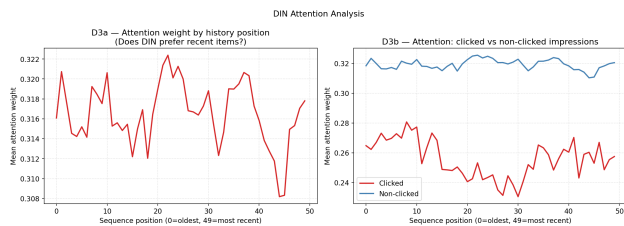


Figure 3. DIN attention weights averaged over 5,000 test impressions across 50 historical positions. Position 1 is the most recent interaction; position 50 is the oldest.

Stratifying by click outcome shows that the attention distribution remains roughly flat for both clicked and non-clicked impressions, with no systematic preference for recent or distant items. The flat pattern observed earlier persists across click outcomes, consistent with DIN’s design intent of attending based on candidate-ad relevance rather than position.

Taken together, these findings confirm that DIN’s attention mechanism favors *relevance over recency*: the model attends to historical behaviors semantically related to the candidate ad regardless of when they occurred, consistent with DIN’s original design intent (Zhou et al., 2018).

7.4. Feature Importance

LightGBM’s split-gain importance (Figure 14) identifies `user_ctr`, `user_cate_ctr`, and `ad_ctr` as the three most important features, accounting for over 60% of total gain. The SHAP values (Figures 15–16) confirm this ranking and add direction: high `user_ctr` strongly increases predicted click probability, while `user_cate_ctr` provides additional signal beyond per-user CTR alone. LR coefficients (Figure 17) tell the same story: aggregate CTR features dominate, demographic features contribute moderately, and high-cardinality ID features have near-zero coefficients. For both classical models, predictive signal comes almost entirely from hand-crafted click statistics rather than raw IDs—which is precisely what these models cannot exploit. The advantage of deep models thus appears to come primarily from their ability to learn distributed embeddings for high-cardinality categorical features, a capacity that no amount of additional data can substitute for in classical methods, consistent with the saturation patterns observed in Section 6.3.

8. Discussion

Our results provide a coherent picture of the mapping between representational capacity and predictive ability in e-commerce CTR prediction. The progression of AUC from LR to DIN is tightly aligned with four stages of architectural advance, with consistent gains across all user segments we evaluated. But the pattern of where these increases are found, and where they begin to plateau, paints a more nuanced picture than the simplistic assumption that more sophisticated models are always better.

Diminishing returns and the constraints of behavioral modeling. The highest boost in performance in our comparison is achieved by incorporating nonlinearity (LR $\rightarrow$ LightGBM, +0.019 AUC), and the second largest by learned ID embeddings (LightGBM $\rightarrow$ Wide & Deep, +0.015 AUC). By comparison, the attention mechanism of DIN only slightly outperforms Wide & Deep and DeepFM, with a gain of around +0.002 in AUC. Our sample complexity analysis in Section 6.3 reveals that the main barrier is not data volume. Instead, it seems that the benefits of attention-based interest modeling are bounded by the structure of the dataset itself. The behavior history is only 8 days long, and 58.8% of users have fewer than 10 impressions, which means the behavior sequences available to DIN are short and sparse. This view is consistent with the flat average attention distribution presented in Section 7.3. For sparse user sequences, the attention mechanism learns weakly varying importance weights and converges to uniform pooling. However, we expect attention-based modeling to yield larger gains in settings with denser user histories dataset.

Classical and deep models capture different users. From the pairwise recall matrix (Section 7.1) we observe a distinct separation between classical baselines and deep models. Models in the same cluster catch significantly more of the same clicks than models across clusters, with within-cluster recall around twice as large as cross-cluster recall (0.132 vs. 0.061 for LR–DIN). This means the two classes of models describe qualitatively different portions of click behavior. One practical implication is that an ensemble of a classical model and a deep model can recover clicks that each model alone misses. Such hybrid approaches may thus achieve larger gains than the continuous expansion of architectural complexity within a single model family.

Cold start as a fundamental bottleneck. Three independent investigations converge on the same conclusion: prediction error is mainly due to user-side cold start. Nearly 60% of all clicks are missed by every model, and these hard clicks come from low-activity users clicking on low-CTR ads. The false-negative rate rises sharply for users with fewer than 10 impressions, and DIN’s AUC falls to 0.56 in the lowest `user_ctr` decile. Importantly, this drop in performance appears to be uniform across all five models, showing that the bottleneck is lack of signal rather than lack of model capacity. Closing this gap will likely involve fundamentally new sources of information (e.g., content-based ad embeddings, cross-session behavioral signals, richer item-side data beyond simple category identifiers).

Open questions. Our results leave open the question of whether model consensus could serve as a useful calibration signal. Impressions classified as positive by all five models are clicked at 14.7% precision—more than three times the global CTR—whereas impressions with zero model agreement reach only 4.1% precision. This suggests that disagreement across diverse models could serve as a useful proxy for predictive uncertainty and might be leveraged for confidence-aware ranking. We defer this direction to future work.

9. Conclusion

We presented a controlled empirical comparison of five CTR prediction models spanning the full spectrum from logistic regression to attention-based deep learning on the Alibaba Taobao Display Advertising dataset. The AUC progression from LR to DIN confirms that each architectural advance contributes measurable gains, with the largest improvements coming from nonlinearity and learned ID embeddings rather than from attention-based user interest modeling. Across all analyses, user-side cold start emerges as the dominant bottleneck: 60% of all clicks are missed by every model come from low-activity users where no architecture, regardless of complexity, achieves meaningful discrimination. This suggests that future progress in e-commerce CTR predic-

tion will depend less on architectural innovation within the current paradigm and more on incorporating qualitatively richer signals for users with limited behavioral history.

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Appendix

A. Tables

Table 2. Overview of the four raw tables: scale and key fields.

Table	Key Fields	Rows
raw_sample	user, time_stamp, adgroup_id, pid, clk	26,557,961
adfeature	adgroup_id, cate_id, campaign_id, customer, brand, price	846,811
user_profile	user, cms_segid, cms_group_id, final_gender_code, age_level, pvalue_level, shopping_level, occupation, new_user_class_level	1,061,768
behavior_log	user, time_stamp, btag, cate, brand	723,268,134

B. Dataset Figures

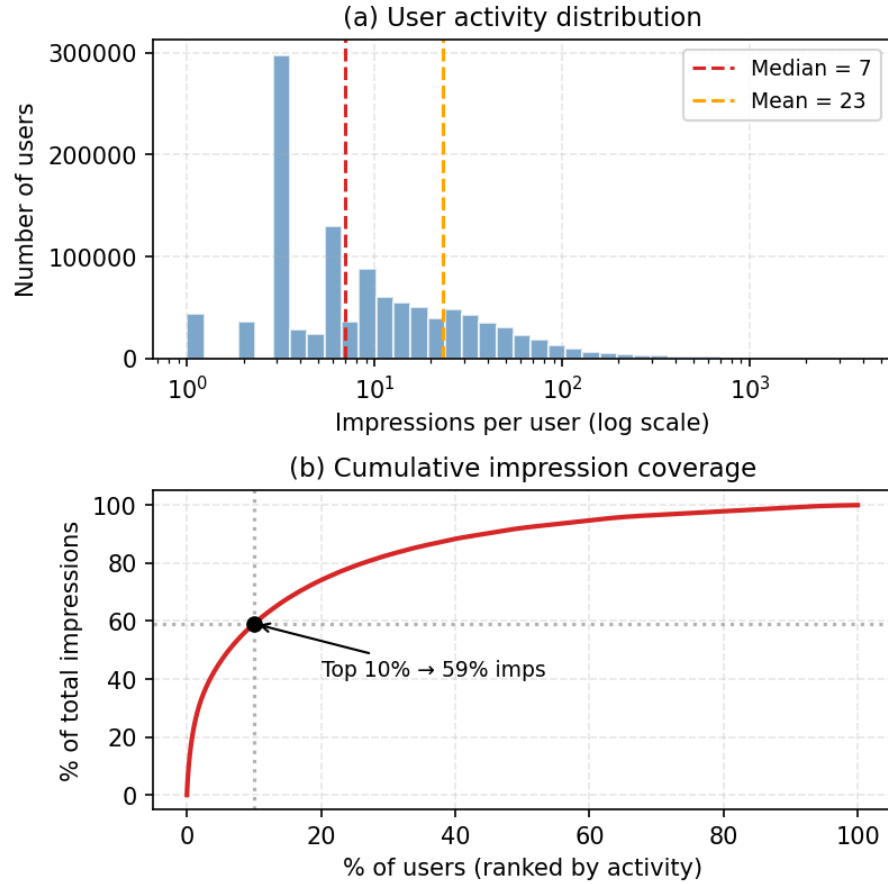


Figure 4. User-activity distribution. (a) Histogram of impressions per user (log x -axis); the median user receives only 7 impressions while the mean is 23.3, evidencing a strong long tail. (b) Cumulative impression coverage; the top 10% of users account for the majority of all impressions.

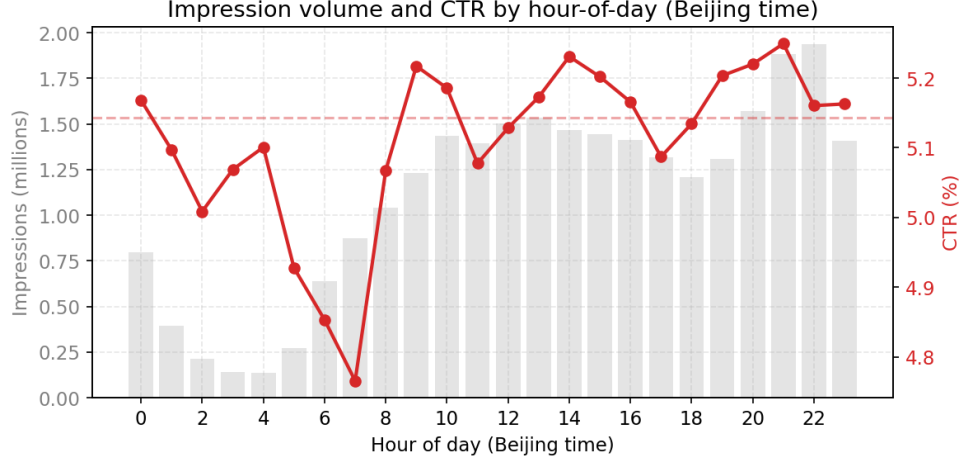


Figure 5. Impression volume (gray bars, left axis) and CTR (red line, right axis) by hour of day in Beijing time. Volume varies $14\times$ across the day while CTR remains within a 0.2 percentage-point band.

C. Datapoint-Level Agreement (D1)

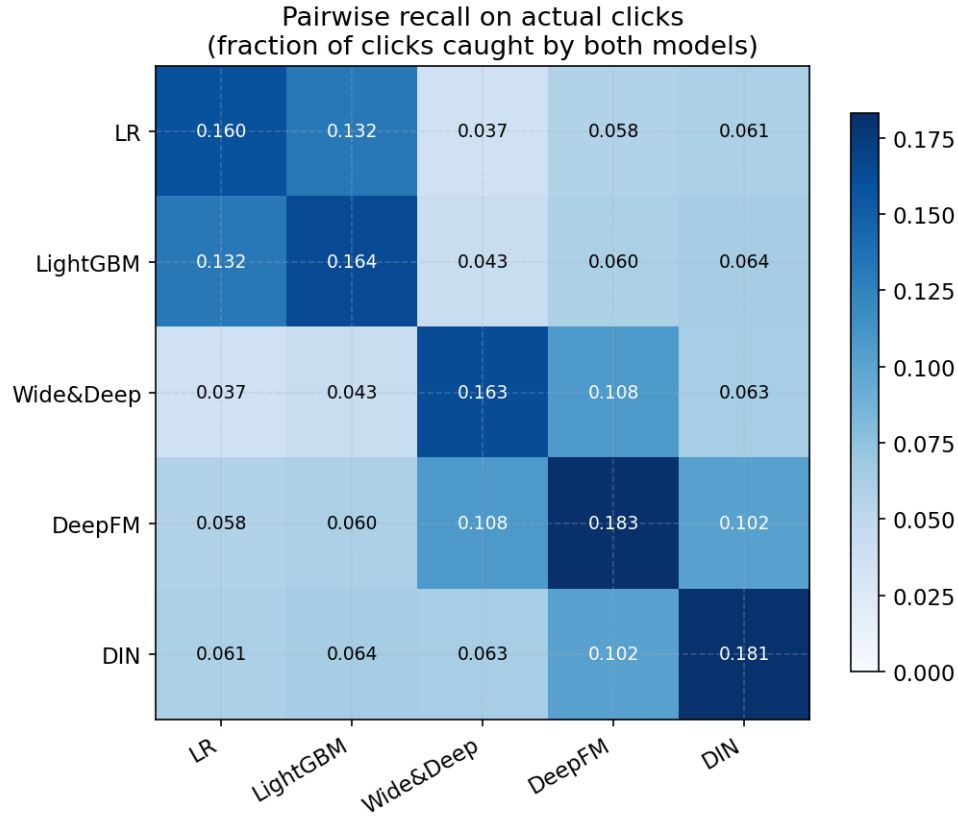


Figure 6. Pairwise agreement matrix on actual clicks. Entry (i, j) is the fraction of true clicks caught by both model i and model j at the top-10% prediction threshold. Diagonal entries are individual recall. Two clusters emerge: classical baselines (LR, LightGBM) have high mutual agreement (0.132); deep models (DeepFM, DIN) form a separate cluster (0.102). Cross-cluster agreement is markedly lower (LR vs. DIN: 0.061), suggesting the two families capture qualitatively different sub-populations of clickers.

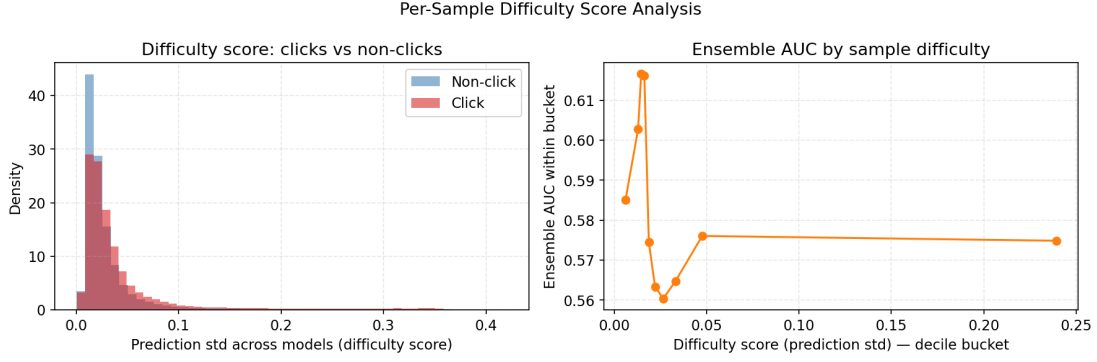


Figure 7. Per-sample difficulty score, defined as the standard deviation of predicted probabilities across the five models. Clicked impressions (red) and non-clicks (blue) are shown separately.

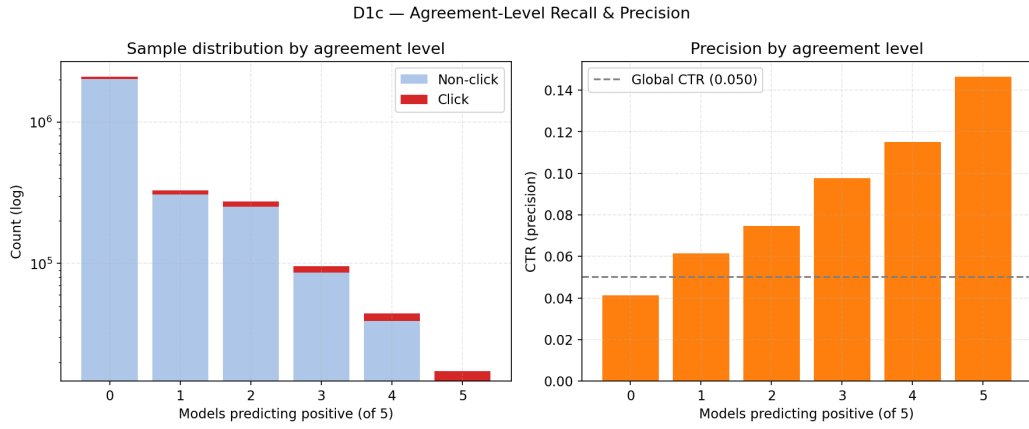


Figure 8. Sample distribution (left) and precision (right) at each agreement level, where the agreement level is the number of models out of five that flag an impression as a click.

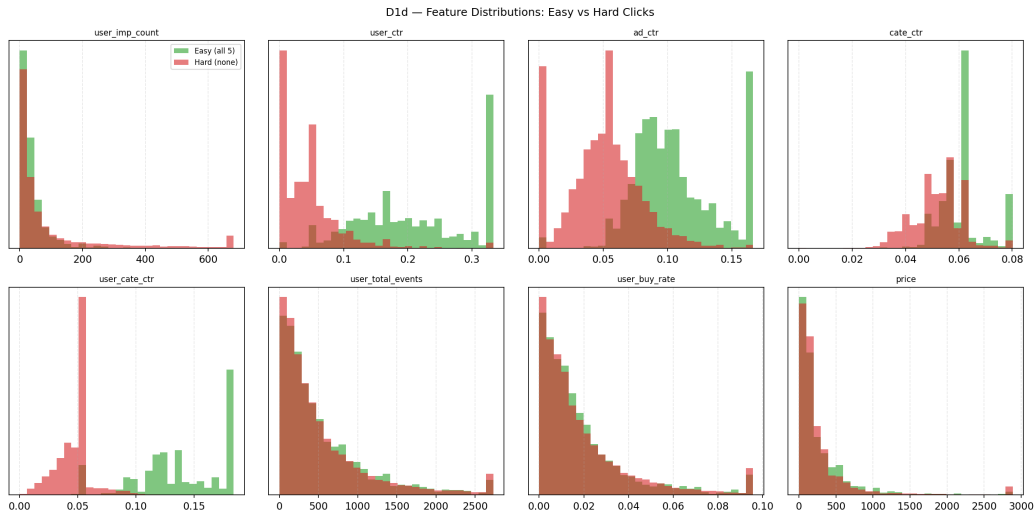


Figure 9. Feature distributions for “easy” clicks (caught by all 5 models) versus “hard” clicks (caught by no model). Easy clicks correspond to high-CTR users browsing high-CTR ads ($\bar{u}_{ctr} = 0.23$, $\bar{a}_{ctr} = 0.117$), while hard clicks (60% of all positives) come from low-activity users on low-CTR ads where neither user-level nor ad-level signal suffices.

D. Failure Mode Analysis (D2)

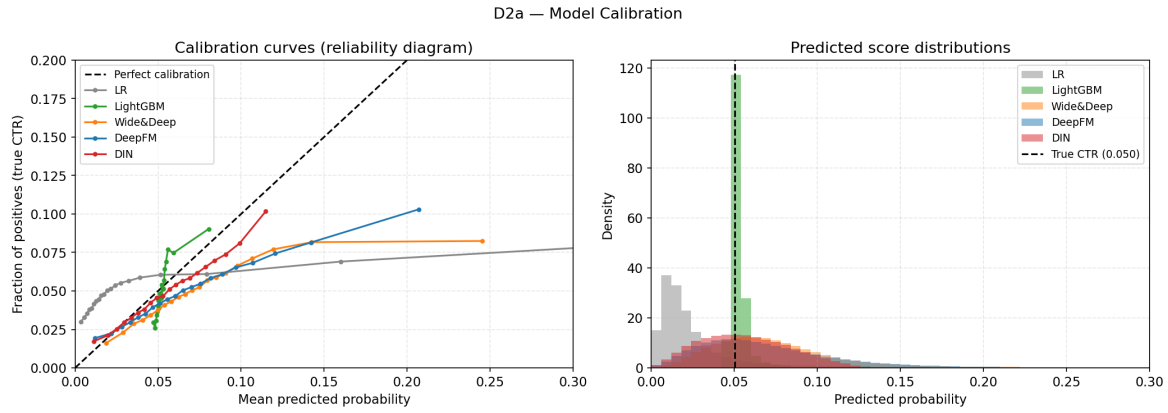


Figure 10. Reliability diagrams (left) and predicted-score distributions (right) for all five models. The dashed diagonal in the left panel marks perfect calibration.

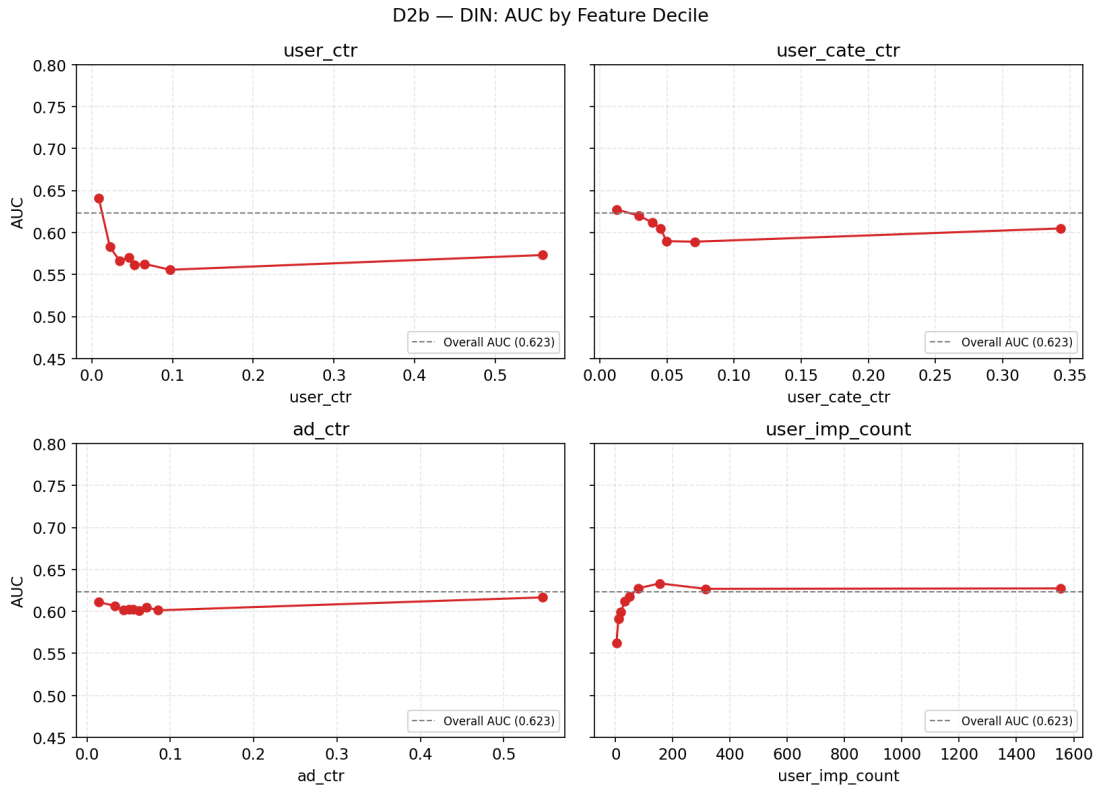


Figure 11. DIN's test-set AUC stratified by deciles of key features. AUC is highest in the top decile of `user_ctr` (around 0.64) and drops to 0.56 in the lowest decile, showing that DIN performs best on high-signal users while its discriminative power weakens in cold-start regimes.

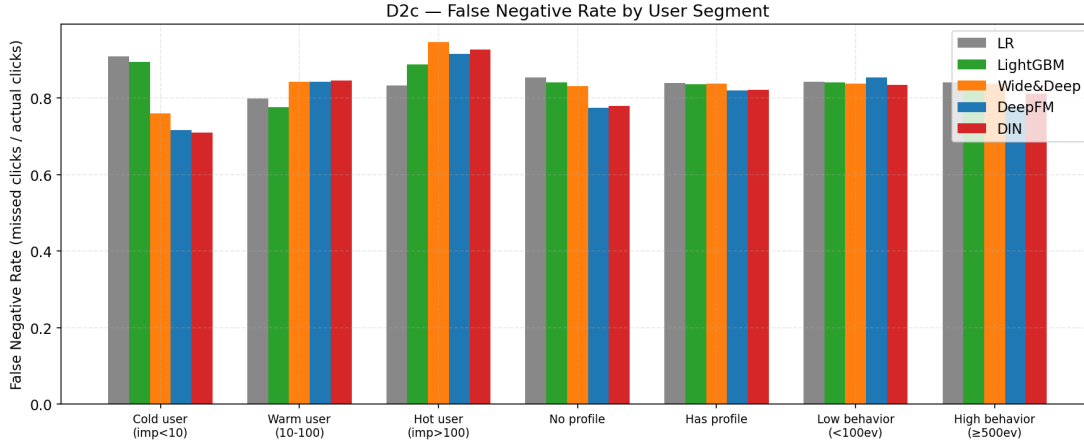


Figure 12. DIN’s false-negative rate (missed clicks) stratified by user activity level. FNR rises sharply for users with fewer than ~ 10 historical impressions, identifying user-side cold start as the dominant residual error source.

E. Per-Segment Performance Comparison

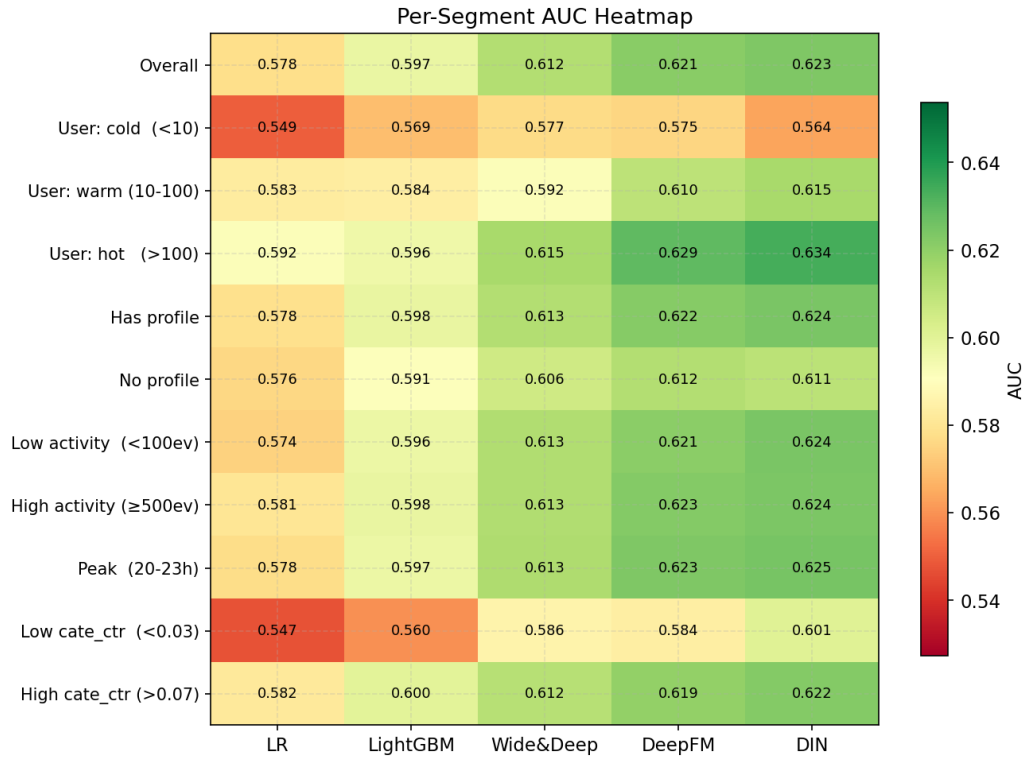


Figure 13. Heatmap of test-set AUC across (model, segment) cells. Rows correspond to the five models; columns correspond to user-activity buckets. The lowest-activity bucket is uniformly weakest across all models, indicating that cold-start users are the clearest target for future modeling improvements rather than a niche where any one architecture struggles disproportionately.

F. Feature Importance (D4)

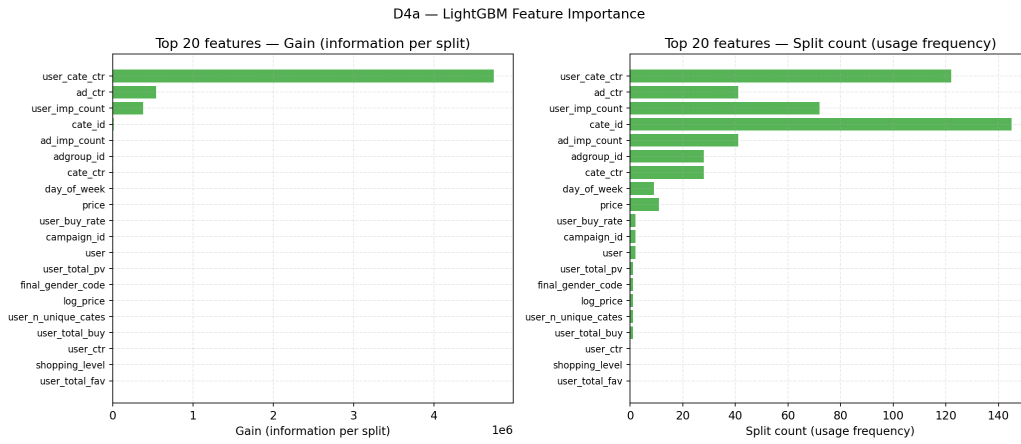


Figure 14. LightGBM top-20 feature importance, measured as gain (information per split, left) and split count (usage frequency, right).

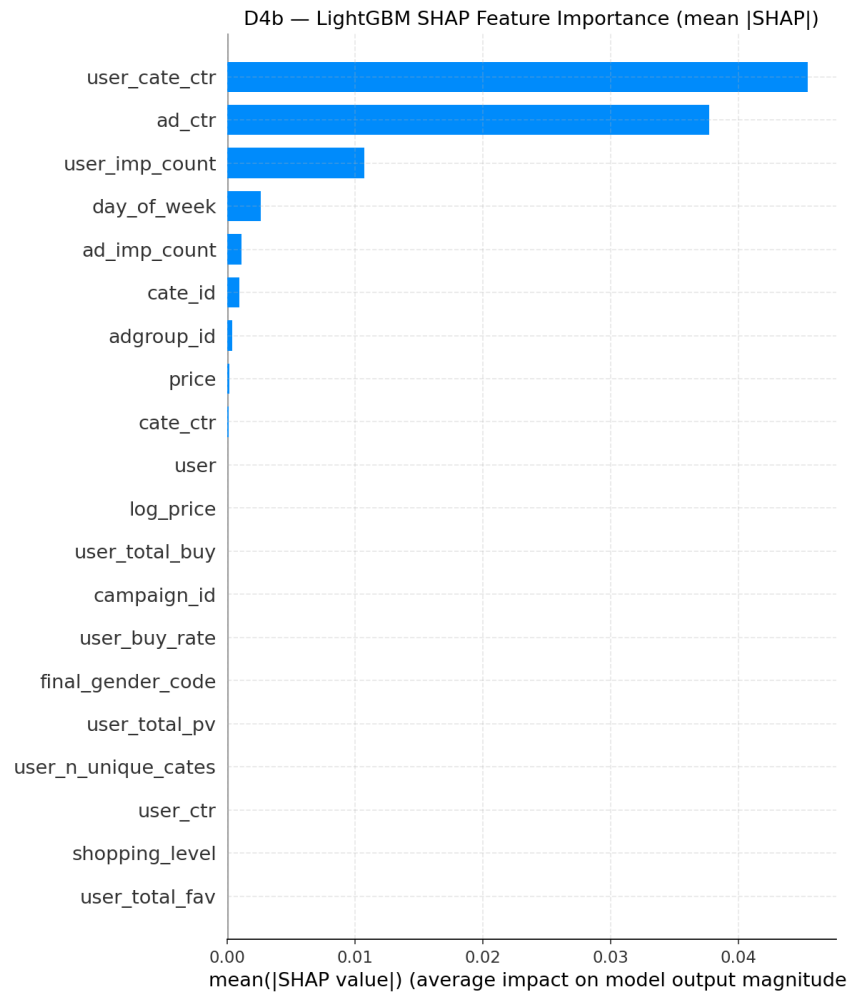


Figure 15. Mean absolute SHAP values for LightGBM, computed on 5,000 test samples. The ranking is consistent with the gain-based importance in Figure 14, confirming that empirical CTR aggregates dominate the model’s predictive signal.

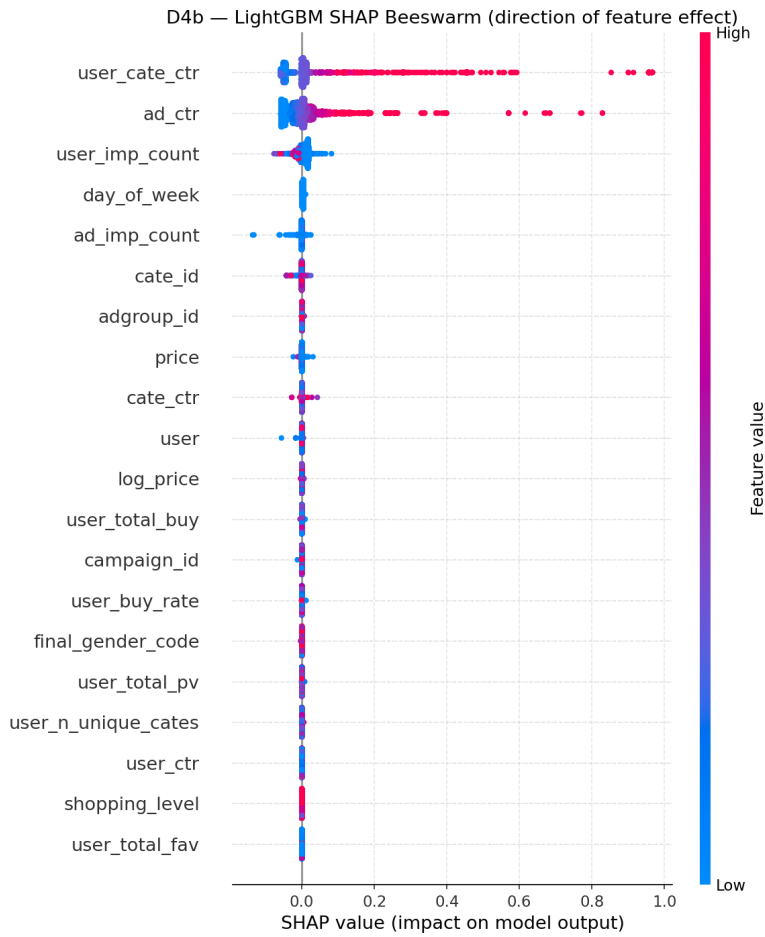


Figure 16. SHAP beeswarm plot showing the per-sample distribution and direction of each feature’s effect on LightGBM predictions. Points are colored by feature value (blue = low, red = high). High user_ctr pushes predictions strongly upward, while user_cate_ctr contributes cross-effect signal beyond per-user CTR alone.

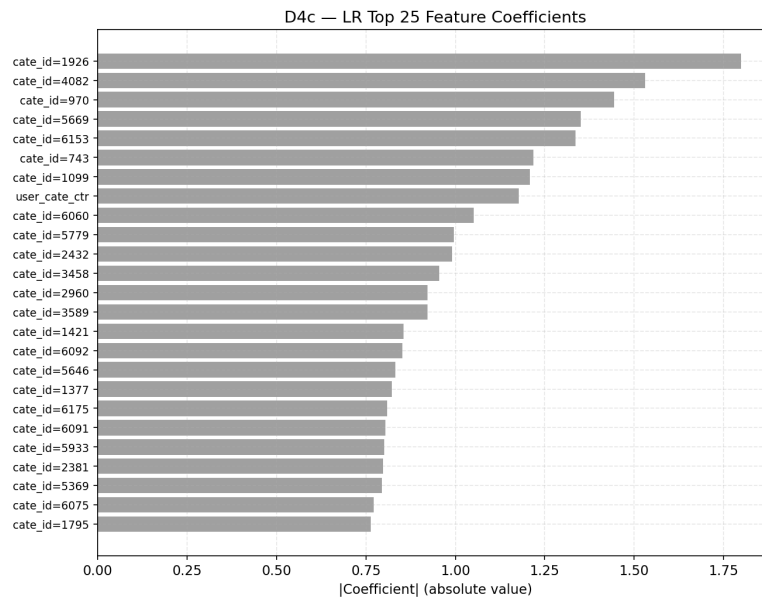


Figure 17. Top 25 features by absolute coefficient magnitude in the logistic regression baseline.